# MAT-Net: Multiscale Aggregation Transformer Network for Hyperspectral Unmixing

Pengrui Wang, Rong Liu, *Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Deep learning (DL) has recently shown considerable potential in the field of Hyperspectral unmixing (HU), thanks to its advanced capabilities in feature representation. Most joint spectral-spatial DL-based unmixing methods extract spatial features at a single scale, with basic feature fusion techniques. Given the heterogeneity in ground object sizes, these methods, with their singular receptive fields, may capture only incomplete features or blend features from disparate objects, leading to inadequate spatial comprehension. To tackle this challenge, we propose the multiscale aggregation transformer network (MAT-Net), which utilizes an encoder–decoder architecture designed to harness both spectral and spatial data comprehensively. Within this structure, we have devised a dual-stream, multibranch convolutional neural network (CNN) encoder to extract both spectral and multiscale spatial information. The spatial stream includes three CNN branches with different receptive fields. A block-by-block branching strategy is employed in this stream to maintain the continuity of multiscale spatial information and reduce computational costs. To effectively integrate intricate multiscale features, we introduce a transformer encoder that incorporates multihead self-scale-aggregation attention (MSsaA) blocks. These blocks are designed to adaptively modulate feature weights based on the scale characteristics of the image, enabling a more nuanced feature integration process. Experimental results on synthetic and real datasets demonstrate the effectiveness of MAT-Net.**

***Index Terms*— Convolutional neural network (CNN), deep learning (DL), hyperspectral unmixing (HU), multihead self-attention mechanism, transformer network.**

## I. Introduction

HYPERSPECTRAL remote sensing images (HSIs) capture data across a broad wavelength range, recording continuous spectral bands. With rich spectral information, HSIs have established considerable utility in various applications like plant disease detection [1], tree species classification [2], and pollution monitoring [3]. However, sensor limitations of HSIs result in a significant presence of mixed pixels, which encapsulate spectral information from multiple materials, posing considerable challenges for pixel-level analyses such as classification [4] and detection [5]. To mitigate the mixed pixel problem, hyperspectral unmixing (HU) emerges as a pivotal technique, effectively decomposing pixels into their constituent pure materials (endmembers) and quantifying their fractional presence (abundances) [6].

The linear mixing model (LMM) is characterized by its simplicity and efficiency compared to non-LMM, thereby establishing its widespread adoption in HU [7]. According to the LMM, mixed pixels are linear combinations of the endmembers contained within, with the abundances serving as the corresponding weights. The LMM incorporates three constraints to ensure a physically meaningful interpretation: the endmember nonnegativity constraint (ENC), the abundance nonnegativity constraint (ANC), and the abundance sum-to-one (STO) constraint (ASC).

HU methods grounded in LMM can be divided into two main categories based on their workflow [8]. The first category follows a two-phase workflow: endmember extraction and abundance estimation. Endmembers are derived from prior knowledge or identified using endmember extraction algorithms (EEAs). EEAs can be divided into endmember identification algorithms (EIAs) and endmember generation algorithms (EGAs). EIAs assume the existence of pure pixels in HSIs and directly extract them from the image. Frequently employed EIAs include convex geometry-based [8], spatial-spectral-based [12], [13], [14], and combinatorial optimization-based algorithms [15], [16], [17]. Vertex component analysis (VCAs) [11], a representative convex geometry-based EIA, extracts endmembers with high efficiency by iteratively projecting the data onto the orthogonal direction of the subspace spanned by the already extracted endmembers. Spatial-spectral-based algorithms combine spatial information to enhance the algorithms' robustness to noise interferences, which prefer pixels located in homogenous regions as endmembers. Combinatorial optimization-based algorithms use evolutionary techniques to find the optimal endmember combination that best fits the objective function. These algorithms can conveniently design and optimize complex nonconvex or multiple objective functions. In contrast, EGAs generate virtual endmembers without requiring pure pixels in the image. Notable examples include minimum volume-based algorithms [18], [19], which obtain endmembers by finding the simplex with the minimum volume enclosing all pixels. Subsequent abundance estimation typically utilizes constrained least squares unmixing [20] or sparse

Received 4 August 2024; revised 20 October 2024; accepted 29 October 2024. Date of publication 11 November 2024; date of current version 20 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62201622; in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515010785; and in part by the Opening Project of Guangdong Province Key Laboratory of Computational Science, Sun Yat-sen University, under Grant 2024019. *(Corresponding author: Rong Liu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3494795

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unmixing [21], [22]. Constrained least squares require accurate endmembers, whereas sparse unmixing selects suitable endmembers from a spectral library, concurrently estimating their abundances.

The other category conducts joint estimation of endmembers and abundances. Common algorithms in this category include statistical-based algorithms [7] and deep learning (DL)-based algorithms [23], [24], [25]. Representative statistical-based algorithms, such as nonnegative matrix factorization (NMF)-based algorithms [26], [27], [28], [29], add a variety of constraints to the abundance matrix or endmember matrix based on the fundamental NMF model to ensure the decomposition results are as close as possible to the optimal solution.

DL has gained traction in remote sensing over recent years due to its advanced feature representation capabilities [30], [31]. The autoencoder (AE) framework is widely utilized in HU, where an encoder transforms the input image into a low-dimensional feature representation. Constraints are applied to satisfy the ANC and ASC to obtain abundances. A decoder then uses the endmember matrix as weights, combining it with the abundance matrix to reconstruct the image.

Early AE-based unmixing algorithms are mainly pixel-level networks utilizing only spectral information, with encoders typically consisting of fully connected layers, 1-D convolutional layers, or $1 \times 1$ 2-D convolutional layers, trained pixel by pixel. Examples include SNSA [32], EndNet [25], DAEN [24], uDAS [33], EGU-Net [34], and CyCU-Net [35]. SNSA utilizes stacked nonnegative sparse AEs to detect outliers and then employs a final AE for unmixing, performing well in HSIs with outliers and low signal-to-noise ratios (SNRs). EndNet enhances the original AE structure with extra layers, and it introduces a novel loss function combining spectral angle distance (SAD) and Kullback–Leibler divergence to improve unmixing. DAEN employs stacked AEs to learn spectral features, providing robust initialization for the variational AE, reducing the impact of outliers. uDAS removes the nonnegativity constraint, incorporates denoising, and $l_{21}$ constraints to address the inherent defects of AEs and avoid additional errors. EGU-Net develops an auxiliary network for unmixing pure pixels extracted by EEAs, guiding the main network training by sharing weights. CyCU-Net employs two cascaded AE networks for cyclic image reconstruction and introduces an abundance consistency constraint to minimize information loss during unmixing. These methods excel at extracting spectral features but do not fully utilize the spatial correlations among adjacent pixels.

To leverage spatial correlations, spatial-level unmixing networks have been developed, employing strategies such as 2-D convolutions and transformers to extract spatial information. For example, CNNAEU [23] uses 2-D Convolutional neural networks (CNNs) to leverage spatial information without pooling layers to retain spatial detail. DeepTrans [36] introduces transformers to extract global spatial information for unmixing.

However, methods focusing only on spectral or spatial information overlook the potential benefits of a comprehensive approach. To overcome this limitation, 3-D convolutional networks have been explored for joint spectral-spatial unmixing [37], [38], but their complexity and parameter number increases. Recently, joint spectral-spatial networks utilizing two streams to extract spectral and spatial features have been proposed. The dual-branch AE (DBA) network [39] extracts spectral features with fully connected layers and spatial features with 2-D convolution. SSANU-Net [40] utilizes a nonlinear decoder with a dual-stream encoder for non-LMM-based unmixing. SSABN [41] incorporates a channel attention mechanism in its spectral stream and constructs its spatial stream using the Swin Transformer block (STB). Most of the spectral-spatial networks extract single-scale spatial features, which are insufficient for representing the complex heterogeneity of ground objects in real-world data. Ground objects in HSIs exhibit varying feature scales, such as the different texture smoothness in vegetation, roofs, and rivers. A single-branch structure in the spectral stream is limited to fixed receptive fields and lacks adaptability.

Some researchers extract multiscale spatial features combined with spectral features to enhance unmixing. Gao et al. [42] proposed a dilated convolution extended-aggregated strategy to obtain multiscale spatial features. HUMSCAN [43] employs a channel attention module for spectral features and a spatial attention module with different kernel sizes for spatial features. MsCM-Net [44] processes data at different scales by using multiple pooling operations, and a regional mask strategy is employed to improve accuracy. However, these methods often use simplistic feature fusion strategies like weighted summation and concatenation. Weighted summation cannot achieve pixel-level adjustment of fusion weights. The concatenation approach first concatenates multiscale features and then uses $1 \times 1$ 2-D convolutional layers for dimensionality reduction. This single-layer convolution lacks sufficient feature expression and precise weight adjustment. Additionally, due to the channel fusion characteristics of the $1 \times 1$ convolution kernel, this method may cause feature contamination between endmembers.

In addition to CNNs, advanced architectures from the field of computer vision (CV) are being integrated into the AE framework for HU, achieving notable results. Examples include generative adversarial networks (GANs) [45], deep image prior (DIP) [46], and recurrent neural networks (RNNs) [47]. Some studies have integrated the AE framework with sparse unmixing or NMF [48], [49]. Additionally, model-driven deep neural networks (DNNs) have been proposed to mitigate the black-box nature of DL [50], [51].

Moreover, Transformers have gained increasing attention in the field of unmixing, with DeepTrans being a representative method. DeepTrans begins by performing patch embedding on spectrally compressed HSI before feeding it into a Transformer encoder. The difference between this encoder and the one used in ViT is that it features a modified attention mechanism where the Query vector is derived solely from the classification token. Finally, the output feature map is reshaped and passed through a decoder to obtain the reconstructed image. Other Transformer-based unmixing approaches adopt similar strategies by incorporating Transformer blocks into AE-based unmixing frameworks. For instance, SSABN introduces STB into the spatial stream of a dual-stream network.

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Similarly, UST-Net designs a U-Net-like network in which an STB replaces the convolution operation after each pooling layer [52].

### A. Motivation

After reviewing the current DL-based unmixing research, we have identified two key points: 1) integrating spectral and spatial information is crucial for effective HU, and 2) there is an urgent need to develop advanced methods for multiscale feature extraction and feature fusion to enhance HU performance. To this end, we design a network to dynamically and adaptively harness multiscale information.

If the encoder generates complex multiscale features, relying on rudimentary fusion methods may not fully exploit this information. Basic methods fail to finely adjust the weights of multiscale features according to regional characteristics, which are essential for accurate abundance estimation. Recent achievements in CV, particularly the use of transformers, have been encouraging [53], [54]. The vision transformer (ViT) [54], which applies the transformer architecture by embedding image patches as tokens and processing them through multiple layers of transformer encoders [55], has demonstrated powerful dynamic feature extraction capabilities through its multihead self-attention mechanism (MHSA).

In this work, we propose utilizing the transformer architecture for multiscale feature fusion. To effectively achieve this, it is essential to fully realize multiscale information interaction. By inputting multiscale features into MHSA, pixel-level feature interaction can be achieved. The MHSA process involves dot-multiplying the query and key matrices, scaling, and applying softmax to yield an attention score matrix representing the degree of correlation between multiscale features. The attention score matrix is then dot-multiplied with the value matrix to inject attention score information into the multiscale features, enabling multiscale information aggregation. The multihead mechanism allows for precise weight adjustment across multiple feature subspaces formed by different image regions. Finally, the multilayer perceptron (MLP) expands and compresses the dimensions of the fused features, further enhancing the model's feature expressive ability. The MLP consists of two fully connected layers: the first layer expands the feature dimensions to increase the model's nonlinear representation capacity, while the second layer compresses the expanded dimensions back to the original size.

However, applying MHSA to multiscale feature fusion, unlike the classical MHSA which takes words or image patches as tokens, involves inputting whole-image scale features as tokens. This significantly increases token dimensions and using the ViT embedding operation would result in memory overflow when data are passed to the linear layer. To handle this, we need to modify the dimensional flow within the model and adjust the internal structure of the MHSA. Therefore, we propose a specifically modified transformer encoder, referred to as multihead self-scale-aggregation attention (MSsaA).

### B. Contributions

This article proposes a novel multiscale aggregation transformer network (MAT-Net) for HU within an AE architecture. The encoder integrates CNNs to extract spectral and multiscale spatial features with a transformer encoder for effective feature fusion. MAT-Net is an end-to-end network designed to dynamically and adaptively harness multiscale information, providing accurate estimation of endmembers and their abundances. The primary contributions of this article are summarized as follows.

1) We propose a novel HU framework that leverages the transformer architecture's ability to effectively fuse multiscale features extracted by CNNs. This framework enhances the network's ability to sense different scales, improving unmixing performance.
2) We introduce a dual-stream, multibranch CNN encoder for multiscale feature extraction, with separate spectral and spatial streams. The spatial stream uses a multibranch architecture to extract information across three distinct receptive fields. A block-by-block branching strategy in the main branch maintains the integral connectivity of multiscale information while minimizing computational complexity.
3) We develop a transformer encoder for feature fusion, incorporating the proposed MSsaA block to analyze multiscale information. MSsaA treats multiscale features as tokens and, by modifying the internal structure of the MHSA mechanism, applies it to multiscale feature fusion for the first time. By employing stacked MSsaA blocks, the encoder continuously compares correlations and self-correlations among multiscale features at the pixel level, effectively capturing the intrinsic attributes of ground objects in HSIs. This process enhances the network's scale-aware capability and elevates the level of feature fusion. Specifically, by adaptively adjusting feature weights, the transformer encoder optimally aligns features to support more precise abundance estimation.

This article is structured as follows: Section II details the proposed MAT-Net architecture. Section III presents the experimental settings and results. Section IV provides a detailed analysis of the algorithm's effectiveness. Finally, Section V summarizes the conclusions.

## II. Methodology

### A. Overall Architecture

Fig. 1 illustrates the overall architecture of the MAT-Net. The network is an AE-based network, encompassing a hybrid encoder that integrates CNN and transformer, along with a decoder.

Let $\mathbf{Y} \in \mathbb{R}^{L\times C\times R}$ denotes the input HSI, where $C \times R$ represents the spatial dimensions encompassing $L$ spectral bands. The encoder $f_E$ is tasked with transforming $\mathbf{Y}$ into a low-dimensional representation $\mathbf{S}' \in \mathbb{R}^{P\times C\times R}$, where $P$, the number of endmembers, is determined by prior knowledge

$$\begin{aligned} \mathbf{S}' &= f_E(\mathbf{Y}) = f_{Et}(f_{Ec}(\mathbf{Y})) \\ f_E &: \mathbb{R}^{L\times C\times R} \rightarrow \mathbb{R}^{P\times C\times R} \end{aligned} \tag{1}$$

where $f_{Ec}$ represents the CNN encoder. It aims at extracting both spectral and multiscale spatial information from the input

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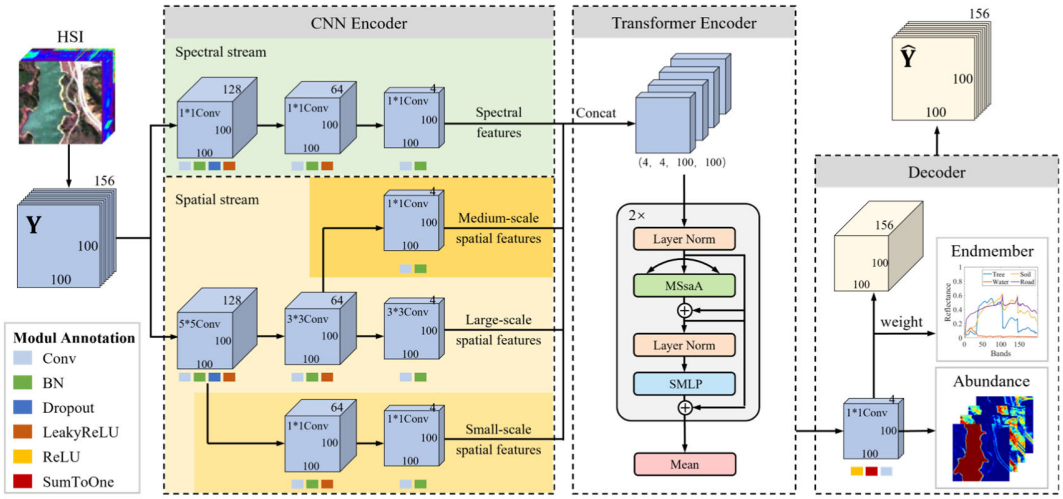



Fig. 1. Overall architecture of MAT-Net.

image. $f_{Et}$ denotes the transformer encoder, which aggregates the multiscale information

$$\mathbf{T} = f_{E_c}(\mathbf{Y})$$
$$f_{E_c} : \mathbb{R}^{L\times C\times R} \rightarrow \mathbb{R}^{4\times P\times C\times R} \tag{2}$$
$$\mathbf{S}' = f_{E_t}(\mathbf{T})$$
$$f_{E_t} : \mathbb{R}^{4\times P\times C\times R} \rightarrow \mathbb{R}^{P\times C\times R} \tag{3}$$

where $\mathbf{T}$ represents the multiscale feature maps. The decoder $f_D$ is used to transform the low-dimensional representation $\mathbf{S}'$ to the reconstructed HSI $\hat{\mathbf{Y}}$

$$\hat{\mathbf{Y}} = f_D(\mathbf{S}')$$
$$f_E : \mathbb{R}^{P\times C\times R} \rightarrow \mathbb{R}^{L\times C\times R}. \tag{4}$$

### *B. Multiscale Feature Extraction Using CNN Encoder*

The proposed dual-stream multibranch CNN encoder can simultaneously extract spectral features and multiscale spatial features through its spectral and spatial streams. The spectral stream comprises three blocks. Each block contains a $1 \times 1$ 2-D convolution (Conv) layer, which progressively compresses the number of bands of the input HSI. The 2-D convolutional layers create new feature maps through a weighted summation over the global bands. Given that the spectral signatures of materials typically span extensive wavelength ranges, employing $1 \times 1$ 2-D convolutional layers is an effective strategy for extracting spectral features from HSIs. To enhance training stability, all three blocks incorporate batch normalization (BN) following the convolutional layers. In the first block, a dropout layer is added after the BN layer to mitigate the vanishing gradient problem in the network. The leaky rectified linear activation function (ReLU) is employed at the output of the first two blocks in the spectral stream to introduce nonlinearity.

The spatial stream of the CNN encoder encompasses three branches, each dedicated to extracting spatial features from the HSIs at large, medium, and small scales, respectively. This design is pivotal, given the variability in object sizes and feature scales within the input image. Effectively leveraging multiscale information is instrumental in enhancing the performance of unmixing methods.

In the spatial stream, the main branch adopts a design similar to the spectral stream, with the key difference being the size of the 2-D convolution kernels utilized. The spatial stream is configured with three blocks, using kernel sizes of $5 \times 5$, $3 \times 3$, and $3 \times 3$, respectively. Since stacking convolutional layers progressively expands the receptive field, these three blocks output spatial features at small, medium, and large scales, respectively.

The main branch is responsible for extracting large-scale spatial features. The output from the second block of the main branch, representing medium-scale spatial features, is fed into the upper lateral branch. This branch consists of a single block that mirrors the design of the spectral stream's final block and employs only $1 \times 1$ 2-D convolutional layers. These layers do not expand the receptive field, thus preserving the feature scale. The $1 \times 1$ convolution reduce the dimensionality of the input features to $P$, simulating the abundance estimation process and decreasing the computational load for subsequent feature fusion. Similarly, the lower lateral branch takes the output from the first block of the main branch to acquire small-scale spatial features.

Since the multiscale information originates from different blocks within the same branch, this block-by-block branching strategy not only ensures the correlation of multiscale features but also reduces computational complexity. The dual-stream multibranch outputs four feature maps, which are stacked into a 4-D matrix $\mathbf{T} \in \mathbb{R}^{4\times P\times C\times R}$, serving as the output of the CNN encoder. The structure of the CNN encoder is shown in Table I.

### *C. Transformer Encoder With MSsaA*

*1) Architecture of the Transformer Encoder:* The transformer encoder consists of two sequentially stacked groups. Each group contains a MSsaA block and a scalewise multilayered perceptron (SMLP) block. Layernorm (LN) is applied

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TABLE I
NETWORK STRUCTURE OF EACH BLOCK WITHIN THE PROPOSED CNN ENCODER

| Blocks | Spectral stream | Spatial stream | | | Unit |
|---|---|---|---|---|---|
| | | Main | Lower | Upper | |
| Block1 | 1×1 Conv<br>BN<br>Dropout<br>Leaky<br>ReLU | 5×5 Conv<br>BN<br>Dropout<br>Leaky<br>ReLU | \ | \ | 128 |
| Block2 | 1×1 Conv<br>BN<br>Leaky<br>ReLU | 3×3 Conv<br>BN<br>Leaky<br>ReLU | 1×1 Conv<br>BN<br>Leaky<br>ReLU | \ | 64 |
| Block3 | 1×1 Conv<br>BN | 3×3 Conv<br>BN | 1×1 Conv<br>BN | 1×1 Conv<br>BN | $P$ |

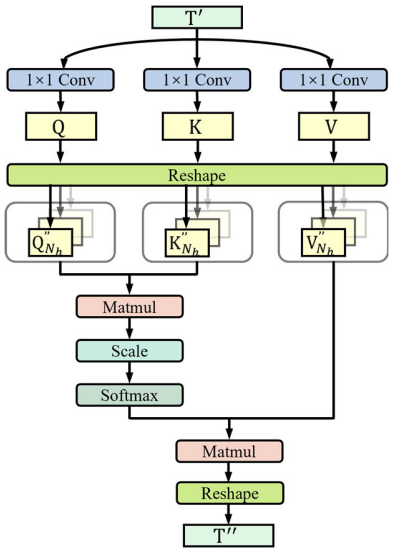



Fig. 2. Architecture of the MSsaA block.

before each block, and residual connections are implemented following every block. The SMLP block fine-tunes the multiscale features using two $1 \times 1$ convolutional layers and a Gaussian error linear unit (GELU) activation function. The overall structure of the transformer encoder can be described as follows:

$$\mathbf{T}_\mathbf{0}' = \mathbf{T} \tag{5}$$

$$\mathbf{T}_\mathbf{i}'' = \mathrm{MSsaA}\left(\mathrm{LN}\left(\mathbf{T}_\mathbf{i-1}'\right)\right) + \mathbf{T}_\mathbf{i-1}', \quad i = 1, 2 \tag{6}$$

$$\mathbf{T}_\mathbf{i}' = \mathrm{SMLP}\left(\mathrm{LN}\left(\mathbf{T}_\mathbf{i}''\right)\right) + \mathbf{T}_\mathbf{i}'', \quad i = 1, 2 \tag{7}$$

$$\mathbf{S}' = \mathrm{mean}\left(\mathrm{LN}\left(\mathbf{T}_\mathbf{2}'\right)\right) \tag{8}$$

where mean() denotes the computation of the mean of the input matrix along the first dimension.

*2) Architecture of the Proposed MSsaA:* Fig. 2 illustrates the architecture of MSsaA block. The details of MSsaA block are introduced step by step as follows:

*Step 1:* The multiscale feature maps $\mathbf{T}$ are directly input as tokens $\mathbf{T}' \in \mathbb{R}^{4\times P\times C\times R}$ into the first MSsaA block. Unlike other transformer-based unmixing networks, our method does not require patch embedding, positional embedding, or the addition of a class token, as there is no sequence information between features of different scales. Due to the excessively large size of each token, we use a $1 \times 1$ convolutional layer instead of a linear layer to obtain $\mathbf{Q}$ (queries), $\mathbf{K}$ (keys), and $\mathbf{V}$ (values)

$$\begin{cases} \mathbf{Q} = f_{\mathrm{Convi}}^{q}\left(\mathbf{W}_i^q\mathbf{T}' + \mathbf{B}_i^q\right) \\ \mathbf{K} = f_{\mathrm{Convi}}^{k}\left(\mathbf{W}_i^k\mathbf{T}' + \mathbf{B}_i^k\right), \quad i = 1, 2 \\ \mathbf{V} = f_{\mathrm{Convi}}^{v}\left(\mathbf{W}_i^v\mathbf{T}' + \mathbf{B}_i^v\right) \end{cases} \tag{9}$$

where $f_{\mathrm{Convi}}^{q}$, $f_{\mathrm{Convi}}^{k}$, and $f_{\mathrm{Convi}}^{v}$ represent the convolutional layers of the two MSsaA blocks; $\mathbf{W}_i^q$, $\mathbf{W}_i^k$, $\mathbf{W}_i^v$, and $\mathbf{B}_i^q$, $\mathbf{B}_i^k$, $\mathbf{B}_i^v$ denote the weights and biases of each convolutional layer, respectively.

*Step 2:* To enhance the network's representation ability, we propose MSsaA for parallel learning across multiple subspaces. First, flatten the multiscale feature maps in queries, keys, and values to obtain $\mathbf{Q}'$, $\mathbf{K}'$, and $\mathbf{V}'$, each with a size of $4 \times D$, where $D = P \times C \times R$. To construct the multihead mechanism, $\mathbf{Q}'$, $\mathbf{K}'$, and $\mathbf{V}'$ are reshaped into $\mathbf{Q}''$, $\mathbf{K}''$, and $\mathbf{V}''$ of size $(N_h \times 4 \times D/N_h)$, where $N_h$ denotes the number of attention heads.

*Step 3:* The self-scale-aggregation attention matrix $\mathbf{A}$ is computed using the attention mechanism. Specifically, the dot product of $\mathbf{Q}''$ and $\mathbf{K}''$ is calculated, then divide by a scaling factor, and a softmax function is applied to get the attention weights. Subsequently, the weight matrix is multiplied by $\mathbf{V}''$ to adaptively utilize the similar scale information from the corresponding regions, resulting in the attention matrix $\mathbf{A} \in \mathbb{R}^{N_h\times 4\times D/N_h}$. The process can be expressed as

$$\mathbf{A} = \mathrm{softmax}\left(\frac{\mathbf{Q}''\mathbf{K}''^{\mathrm{T}}}{\sqrt{D/N_h}}\right)\mathbf{V}'' \tag{10}$$

where $\sqrt{D/N_h}$ is the scaling factor used to mitigate the impact of the dimensions of input tokens on the attention weights.

*Step 4:* To maintain the consistency of feature dimensions, $\mathbf{A}$ is reshaped into a matrix of size $(4 \times P \times C \times R)$ to obtain $\mathbf{T}''$.

### D. Unmixing With Decoder

The result of transformer encoder is $\mathbf{S}' = [\mathbf{s}_1', \mathbf{s}_2', \ldots, \mathbf{s}_P'] \in \mathbb{R}^{P\times C\times R}$. In the decoder, the ReLU activation function and the sum-to-one (STO) operation are applied to ensure that the estimated abundance map $\hat{S}$ satisfies the ANC and ASC, respectively,

$$\hat{\mathbf{S}} = \mathrm{STO}(\mathrm{ReLU}(\mathbf{S}')) \tag{11}$$

where STO() is a normalization operation calculated as

$$\mathrm{STO}\left(\mathrm{ReLU}(\mathbf{S}')\right) = \mathrm{Concat}(\mathrm{norm}_1, \mathrm{norm}_2, \ldots, \mathrm{norm}_P) \tag{12}$$

$$\mathrm{norm}_i = \frac{\mathbf{s}_i'}{\sum_{i=1}^{P}\mathbf{s}_i' + \delta}, \quad i = 1, 2, \ldots, P \tag{13}$$

where $\delta$ is a very small value, and $\mathbf{s}_i'$ represents the feature map of the $i$th endmember contained in ReLU($\mathbf{S}'$).

Finally, the estimated abundance $\hat{\mathbf{S}}$ is passed through a convolutional layer to transform it into the reconstructed HSI $\hat{\mathbf{Y}}$. This convolutional layer is designed without bias, and its weights are initialized using endmembers extracted by VCA [11], a representative convex geometry-based EIA. After training, the weights of the convolutional layer are output as the estimated endmember matrix $\hat{\mathbf{E}} \in \mathbb{R}^{L\times P}$.

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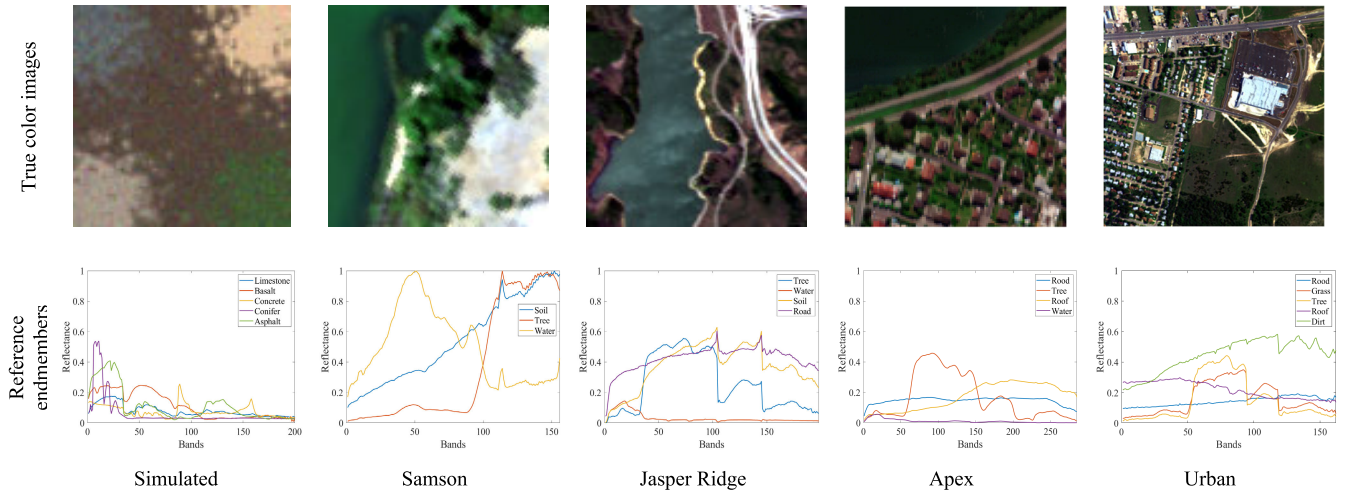



Fig. 3. True color images of the experimental datasets and their reference endmembers.

### E. Loss Function

The SAD loss function was applied to train the proposed network

$$L(\mathbf{Y},\hat{\mathbf{Y}}) = \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{\langle \mathbf{Y}_i, \hat{\mathbf{Y}}_i^T\rangle}{\|\mathbf{y}_i\|_2\|\hat{\mathbf{y}}_i\|_2}\right) \tag{14}$$

where $N$ represents the number of pixels in the input HSI, and $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$ denotes the spectral vector of the $i$th pixel in the original image $\mathbf{Y}$ and reconstructed image $\hat{\mathbf{Y}}$, respectively.

## III. Experiments

### A. Datasets

One simulated HSI and four real HSIs are used to verify the performance of the proposed algorithm. True color images of the datasets and their reference endmembers are displayed in Fig. 3.

*1) Simulated Dataset [56]:* The simulated image contains 60 × 60 pixels with 200 bands. The dataset includes five endmembers: limestone, basalt, concrete, conifer, and asphalt. The spectra for these endmembers were chosen from the ASTER spectral library. The abundance distribution is generated by the Dirichlet distribution.

*2) Samson Dataset [57]:* The Samson image, obtained by the SAMSON sensor, contains 952 × 952 pixels and captures 156 bands covering wavelengths from 401 to 889 nm. The cropped image used in the experiments contains 95 × 95 pixels. Three endmembers exist in the Samson dataset: soil, tree, and water.

*3) Jasper Ridge Dataset [58]:* The Jasper Ridge image, captured by the AVIRIS sensor, consists of 512 × 614 pixels and records 224 bands covering wavelengths from 380 to 2500 nm. The applied subimage, after removing the water vapor and atmospheric effects bands, contains 100 × 100 pixels with 198 bands. Four endmembers exist in this image: soil, water, tree, and road.

*4) Apex Dataset [36]:* The Apex image, obtained by the APEX sensor, contains 110 × 110 pixels and records 285 bands covering wavelengths from 413 to 2421 nm. Four endmembers exist in this scene: water, tree, road, and roof. Due to the unavailability of reference endmembers and their corresponding abundances within the Apex dataset, the selection of reference endmember spectra was manually conducted directly from the image. The reference abundances for this dataset were then estimated using the sparse unmixing by variable splitting and augmented Lagrangian (SUnSAL) method [59].

*5) Urban Dataset [60]:* The Urban image, collected by the HYDICE sensor, consists of 307 × 307 pixels and includes 210 bands ranging from 400 to 2500 nm. For testing hyperparameters of MAT-Net, a cropped image of 300 × 300 pixels was used. After accounting for water vapor and atmospheric interference, 162 bands remain for analysis. Five endmembers exist in this image: asphalt, grass, tree, roof, and dirt.

### B. Evaluation Metrics

The root-mean-square error (RMSE) between the reference abundances $\mathbf{S}$ and the estimated abundances $\hat{\mathbf{S}}$ is used to evaluate the performance of unmixing methods in terms of abundance estimation. The RMSE is calculated by

$$\mathrm{RMSE}(\mathbf{S},\hat{\mathbf{S}}) = \frac{1}{P}\sum_{i=1}^{P}\sqrt{\frac{1}{N}\|\mathbf{s}_i - \hat{\mathbf{s}}_i\|^2} \tag{15}$$

where $\mathbf{s}_i$ is the $i$th reference abundance, and $\hat{\mathbf{s}}_i$ is the corresponding estimated abundance.

The SAD between the reference endmembers $\mathbf{E}$ and the estimated endmembers $\hat{\mathbf{E}}$ is used to evaluate the performance of unmixing methods in terms of endmember estimation. SAD is calculated as follows:

$$\mathrm{SAD}(\mathbf{E},\hat{\mathbf{E}}) = \frac{1}{P}\sum_{i=1}^{P}\arccos\left(\frac{\mathbf{e}_i^{\mathrm{T}}\hat{\mathbf{e}}_i}{\|\mathbf{e}_i\|\|\hat{\mathbf{e}}_i\|}\right) \tag{16}$$

where $\mathbf{e}_i$ represents the $i$th column of the $\mathbf{E}$, and $\hat{\mathbf{e}}_i$ denotes the $i$th column of the $\hat{\mathbf{E}}$.

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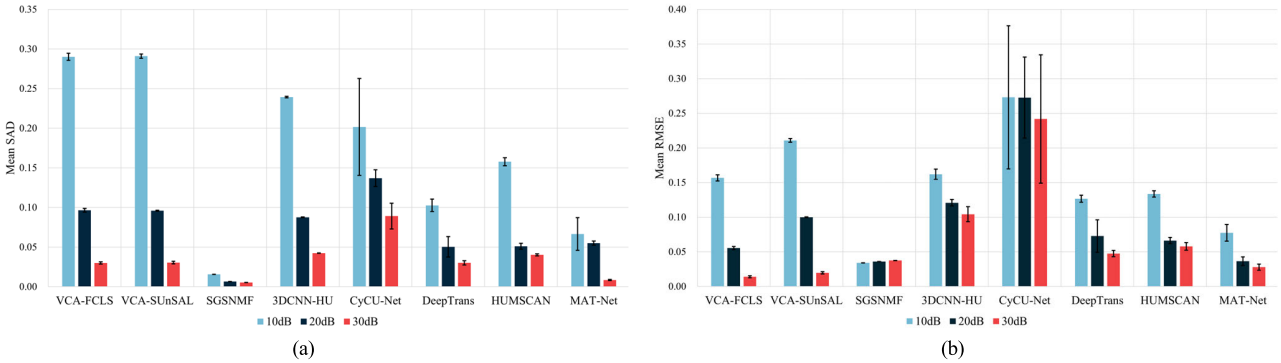



Fig. 4. Experimental results on simulated dataset with different SNRs. (a) SAD. (b) RMSE.

TABLE II
NETWORK HYPERPARAMETER SETTINGS

| | Simulated | Samson | Jasper | Apex | Urban |
|---|---|---|---|---|---|
| $N_h$ | 10 | 15 | 10 | 10 | 10 |
| Learning rate | $2 \times 10^{-3}$ | $9 \times 10^{-3}$ | $9 \times 10^{-3}$ | $9 \times 10^{-2}$ | $5 \times 10^{-3}$ |
| Weight decay | $5 \times 10^{-5}$ | $4 \times 10^{-5}$ | $4 \times 10^{-5}$ | $5 \times 10^{-5}$ | $5 \times 10^{-5}$ |
| Epoch | 400 | 200 | 200 | 200 | 300 |

### C. Experimental Setup

*1) Comparison Algorithms:* The effectiveness of the proposed method is evaluated and compared with seven representative or state-of-the-art unmixing methods from various categories. These include VCA [11] and fully constrained least squares (FCLSs) [20], which represent convex geometry-based and least squares methods, respectively; spatial group sparsity regularized nonnegative matrix factorization (SGSNMF) [49], a notable statistical unmixing method; SUnSAL [59], a prominent sparse unmixing method; and CyCU-Net [35], 3DCNN-HU [61], DeepTrans [36], and HUMSCAN [43], state-of-the-art DL-based unmixing methods. Specifically, CyCU-Net focuses on spatial information extraction, 3DCNN-HU utilizes 3-D convolution for spatial-spectral feature extraction, DeepTrans is the first transformer-based unmixing network, and HUMSCAN emphasizes multiscale spatial features. The initial endmembers for all comparison methods are extracted by VCA.

*2) Hyperparameter Settings:* The hyperparameters for MAT-Net are detailed below: the number of attention heads $N_h$ is a required hyperparameter. The momentum for the BN layer is set to 0.9. Network parameters are updated using the Adam optimizer, with distinct initial learning rates for each dataset. To ensure stable convergence during training, the learning rate is reduced by a factor of 0.8 every 15 epochs. The optimization function includes a weight decay rate, and Kaiming initialization is employed for training the network. Detailed settings for each dataset are provided in Table II. The above parameters are empirical values derived from extensive experimentation conducted in this study.

### D. Experimental Results

*1) Simulated Dataset:* To investigate the robustness of the proposed method against noise, white Gaussian noise with SNRs of 30, 40, and 50 dB was added to the simulated data. The mean and standard deviation of the SAD and RMSE values, calculated from five executions of each method, are displayed in Fig. 4. From the comprehensive results of RMSE and SAD, the proposed method achieves the second-best performance, just after SGSNMF. In high-noise scenarios (10 dB), MAT-Net outperforms all other DL-based methods in both endmember extraction and abundance estimation. While FCLS and SUnSAL yield optimal results in abundance estimation in low-noise scenarios, it's crucial to recognize that these approaches directly extract endmembers from noisy data. In high-noise scenarios, this direct extraction can lead to inaccuracies in endmember identification, significantly affecting the precision of abundance estimation. Overall, our method outperforms other DL-based methods on this dataset.

*2) Samson Dataset:* Table III presents the quantitative evaluation results of all methods on the Samson dataset. MAT-Net exhibits superior performance compared to other methods, particularly in terms of abundance estimation across all endmember classes. The mean RMSE value has significantly improved compared to the suboptimal result of DeepTrans. 3DCNN-HU demonstrates less capability in endmember extraction. SUnSAL and FCLS used the same set of endmembers for abundance estimation, with SUnSAL achieving better performance than VCA. FCLS yields the least favorable results among all compared methods. Inaccurate extraction of endmembers in the water category leads to significant errors in the corresponding abundance obtained by FCLS. Conversely, the SUnSAL method is less impacted by inaccurate endmember extraction.

Fig. 5 depicts the abundance results, showing that the abundance maps produced by the proposed method closely align with the reference abundances across all methods. This alignment is due to the method's adept utilization of multiscale spatial information, which effectively captures the nuanced characteristics of fragmented plots. FCLS, SGSNMF, and 3DCNN-HU underestimate the abundance of soil and trees, leading to water abundance maps saturated with unnecessary details. While SUnSAL shows good visual performance, it struggles with accurate abundance estimation at the boundaries of regions for each ground object. Additionally, DeepTrans and HUMSCAN tend to overestimate water abundance in certain tree pixels. The abundance maps generated by CyCU-Net lacked the detail and smoothness

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TABLE III

QUANTITATIVE COMPARISONS ON THE SAMSON DATASET. THE BEST PERFORMANCES ARE SHOWN IN BOLD, AND THE SECOND-BEST PERFORMANCES ARE SHOWN WITH AN UNDERLINE

| Algorithms | | VCA-FCLS | VCA-SUnSAL | SGSNMF | 3DCNN-HU | CyCU-Net | DeepTrans | HUMSCAN | MAT-Net |
|---|---|---|---|---|---|---|---|---|---|
| SAD | Soil | 0.0404 | 0.0404 | 0.0098 | 0.0346 | 0.0389 | **0.0080** | 0.0156 | 0.0123 |
| | Tree | **0.0219** | **0.0219** | 0.0459 | 0.0588 | 0.0378 | 0.0349 | 0.0353 | 0.0315 |
| | Water | 0.1140 | 0.1140 | 0.2297 | 0.3068 | 0.1503 | 0.0531 | 0.0871 | **0.0390** |
| | Mean | 0.0588 | 0.0588 | 0.0951 | 0.1334 | 0.0757 | 0.0320 | 0.0460 | **0.0276** |
| RMSE | Soil | 0.2693 | 0.1073 | 0.1786 | 0.2284 | 0.1955 | 0.1525 | 0.1812 | **0.0629** |
| | Tree | 0.2508 | 0.0825 | 0.2449 | 0.1929 | 0.1543 | 0.1060 | 0.1726 | **0.0478** |
| | Water | 0.4273 | 0.1395 | 0.3576 | 0.3231 | 0.1851 | 0.2183 | 0.2976 | **0.0532** |
| | Mean | 0.3158 | 0.1098 | 0.2603 | 0.2481 | 0.1783 | 0.1589 | 0.2171 | **0.0546** |

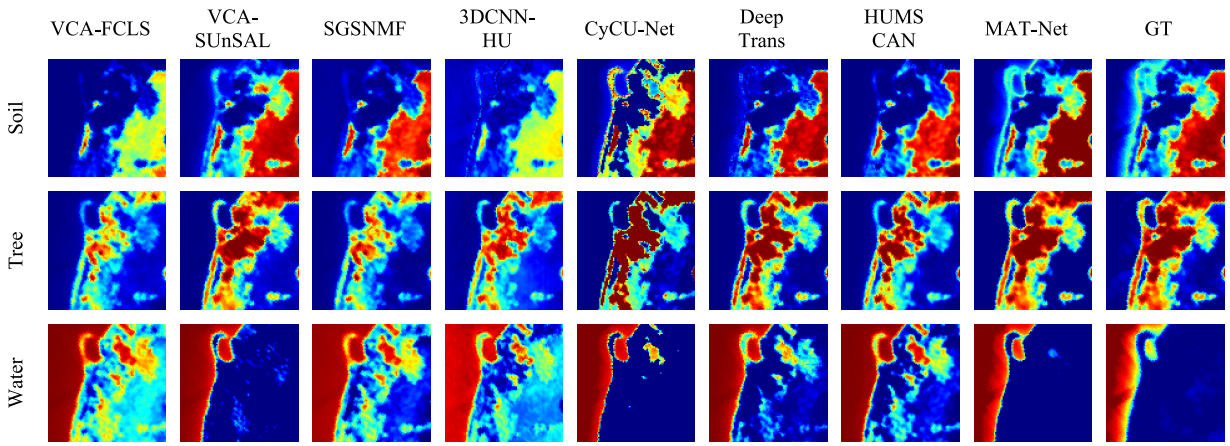



Fig. 5. Visual comparison of the abundances obtained by different methods on the Samson dataset.

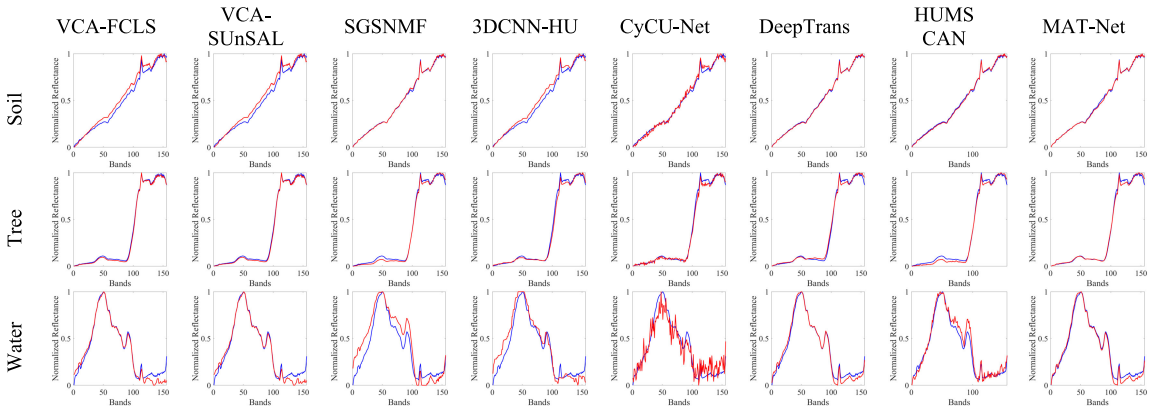



Fig. 6. Visual comparison of the endmembers obtained by different methods on the Samson dataset. Red line: estimated endmembers and blue line: reference endmembers.

present in the reference maps. Fig. 6 presents the estimated endmembers, with those obtained by the proposed method being almost identical to the reference endmembers.

*3) Jasper Ridge Dataset:* Table IV details the quantitative evaluation results of all methods on the Jasper Ridge dataset. It is encouraging to observe that the proposed method surpasses others in terms of SAD and RMSE metrics. MAT-Net has demonstrated outstanding performance in accurately estimating the abundance and endmembers for the soil category.

Figs. 7 and 8 provide visual comparisons of the unmixing results from various methods. The abundance maps and endmember spectra generated by MAT-Net show a closer resemblance to the reference data than those from other methods. The three traditional unmixing methods and the DL-based method 3DCNN-HU tend to overestimate water abundance, while underestimating the abundances of tree and road. Furthermore, the soil abundance maps obtained by these methods lack detail, especially for 3DCNN-HU. All methods, except for CyCU-Net, overestimate the abundance of water in road-dominated pixels. DeepTrans and HUMSCAN both underestimate the abundance of water bodies to varying degrees, leading to water abundance maps lacking certain details.

*4) Apex Dataset:* Table V presents the quantitative evaluation results of all methods on the Apex dataset. The proposed method stands out by achieving the lowest mean values for SAD and RMSE, indicating its superior performance in endmember and abundance estimation compared to other methods.

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TABLE IV

QUANTITATIVE COMPARISONS ON THE JASPER RIDGE DATASET. THE BEST PERFORMANCES ARE SHOWN IN BOLD, AND THE SECOND BEST PERFORMANCES ARE SHOWN WITH AN UNDERLINE

| Algorithms | | VCA-FCLS | VCA-SUnSAL | SGSNMF | 3DCNN-HU | CyCU-Net | DeepTrans | HUMSCAN | MAT-Net |
|---|---|---|---|---|---|---|---|---|---|
| SAD | Tree | 0.1559 | 0.1559 | 0.1366 | 0.1651 | 0.0541 | 0.0768 | **0.0511** | 0.0961 |
| | Water | 0.1219 | 0.1219 | 0.1353 | 0.1679 | 0.0692 | 0.0727 | 0.0812 | **0.0478** |
| | Soil | 0.2453 | 0.2453 | 0.2087 | 0.2156 | 0.1630 | 0.1033 | 0.0565 | **0.0324** |
| | Road | 0.1069 | 0.1069 | **0.0392** | 0.0897 | 0.1036 | 0.0666 | 0.0596 | 0.0508 |
| | Mean | 0.1575 | 0.1575 | 0.1299 | 0.1596 | 0.0975 | 0.0799 | 0.0621 | **0.0568** |
| RMSE | Tree | 0.1640 | 0.1260 | 0.1349 | 0.1536 | 0.1005 | **0.0760** | 0.1105 | 0.0904 |
| | Water | 0.1981 | 0.1614 | 0.1731 | 0.2080 | 0.1454 | **0.1084** | 0.1108 | 0.1271 |
| | Soil | 0.1260 | 0.1228 | 0.1240 | 0.1625 | 0.1017 | 0.1134 | 0.1410 | **0.0369** |
| | Road | 0.1245 | 0.1343 | 0.1135 | 0.1564 | 0.1176 | 0.1009 | 0.1272 | **0.0762** |
| | Mean | 0.1531 | 0.1361 | 0.1364 | 0.1701 | 0.1163 | 0.0997 | 0.1224 | **0.0827** |

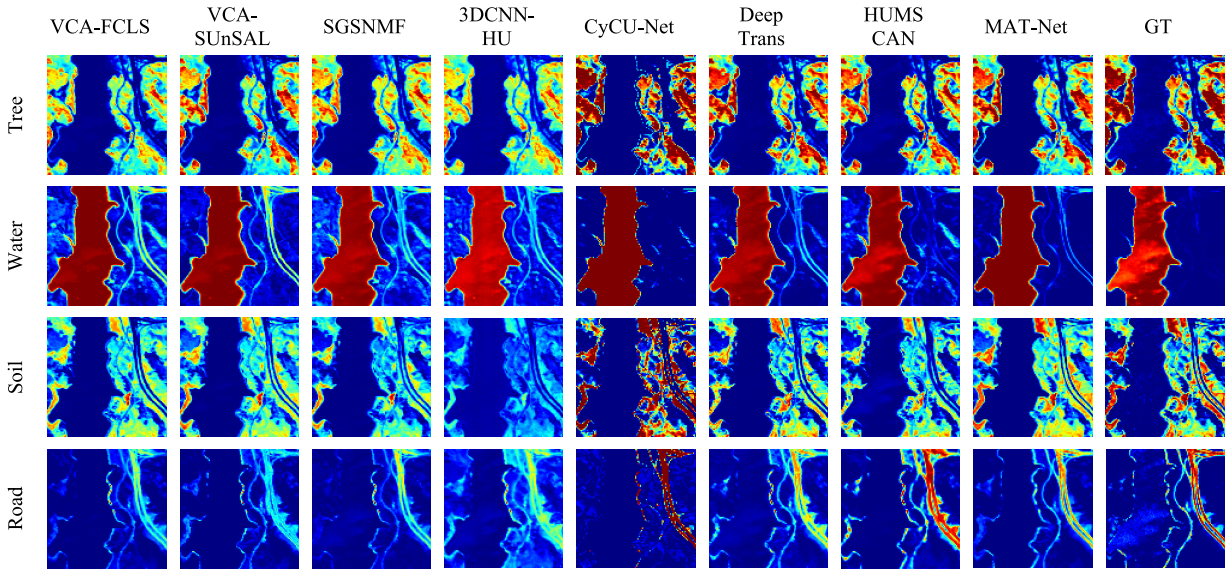



Fig. 7. Visual comparison of the abundances obtained by different methods on the Jasper Ridge dataset.

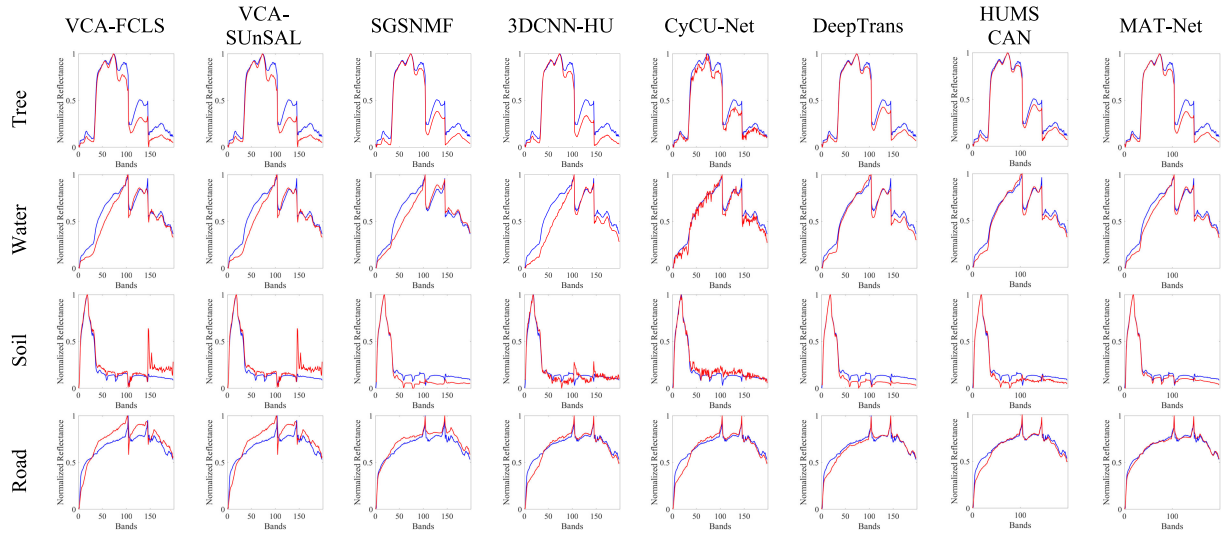



Fig. 8. Visual comparison of the endmembers obtained by different methods on the Jasper Ridge dataset. Red line: estimated endmembers and blue line: reference endmembers.

Compared to FCLS, SUnSAL offers better performance in abundance estimation using the same endmembers extracted by VCA.

From abundance maps displayed in Fig. 9, the results of MAT-Net are visually closest to the reference maps. FCLS, SGSNMF, and 3DCNN-HU substantially overestimate

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TABLE V

QUANTITATIVE COMPARISONS ON THE APEX DATASET. THE BEST PERFORMANCES ARE SHOWN IN BOLD, AND THE SECOND BEST PERFORMANCES ARE SHOWN WITH AN UNDERLINE

| Algorithms | | VCA-FCLS | VCA-SUnSAL | SGSNMF | 3DCNN-HU | CyCU-Net | DeepTrans | HUMSCAN | MAT-Net |
|---|---|---|---|---|---|---|---|---|---|
| SAD | Rood | 0.1001 | 0.1001 | 0.5193 | **0.0668** | 0.1723 | 0.1371 | 0.1392 | 0.0997 |
| | Tree | 0.1275 | 0.1275 | 0.1477 | 0.0909 | 0.1562 | 0.1391 | 0.1314 | **0.0724** |
| | Roof | 0.0964 | 0.0964 | 0.1023 | 0.1031 | **0.0750** | 0.0807 | 0.0978 | 0.0951 |
| | Water | 0.1548 | 0.1548 | 0.2218 | 0.5328 | 0.5981 | **0.0405** | 0.1183 | 0.0477 |
| | Mean | 0.1197 | 0.1197 | 0.2478 | 0.1984 | 0.2504 | 0.0994 | 0.1217 | **0.0787** |
| RMSE | Rood | 0.2123 | 0.1481 | 0.2251 | 0.2066 | 0.2204 | 0.1380 | 0.1355 | **0.0933** |
| | Tree | 0.2306 | 0.1524 | 0.1888 | 0.2361 | 0.1608 | **0.1029** | 0.1463 | 0.1201 |
| | Roof | 0.2213 | 0.1604 | 0.2097 | 0.1868 | 0.2467 | 0.1531 | 0.1092 | **0.1045** |
| | Water | 0.4653 | 0.2842 | 0.4089 | 0.3800 | 0.2749 | 0.1298 | 0.2551 | **0.1225** |
| | Mean | 0.2824 | 0.1863 | 0.2581 | 0.2524 | 0.2257 | 0.1310 | 0.1615 | **0.1101** |

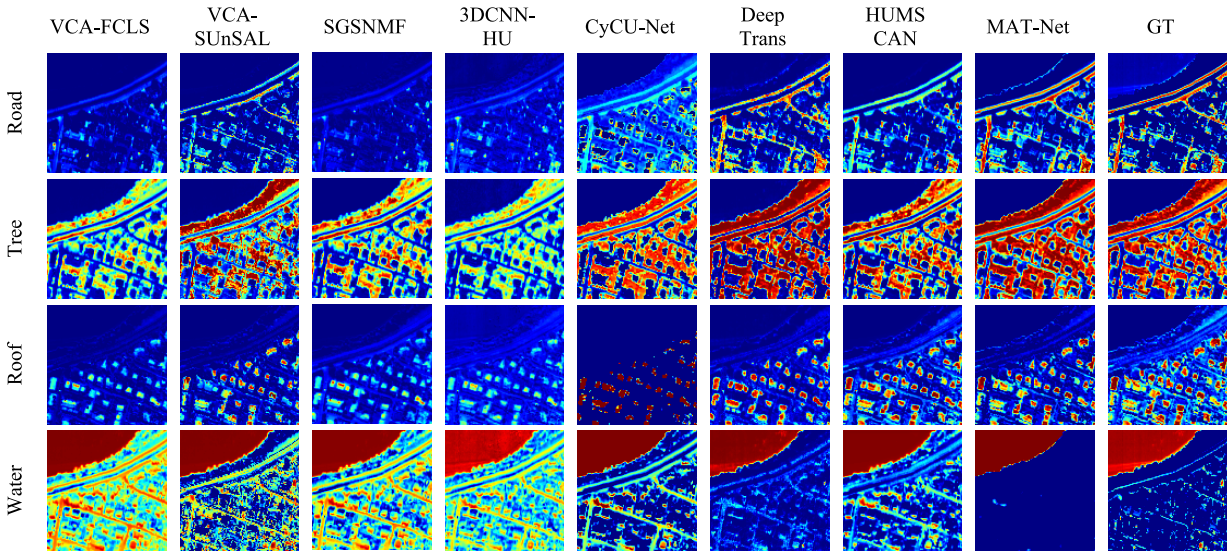



Fig. 9. Visual comparison of the abundances obtained by different methods on the Apex dataset.

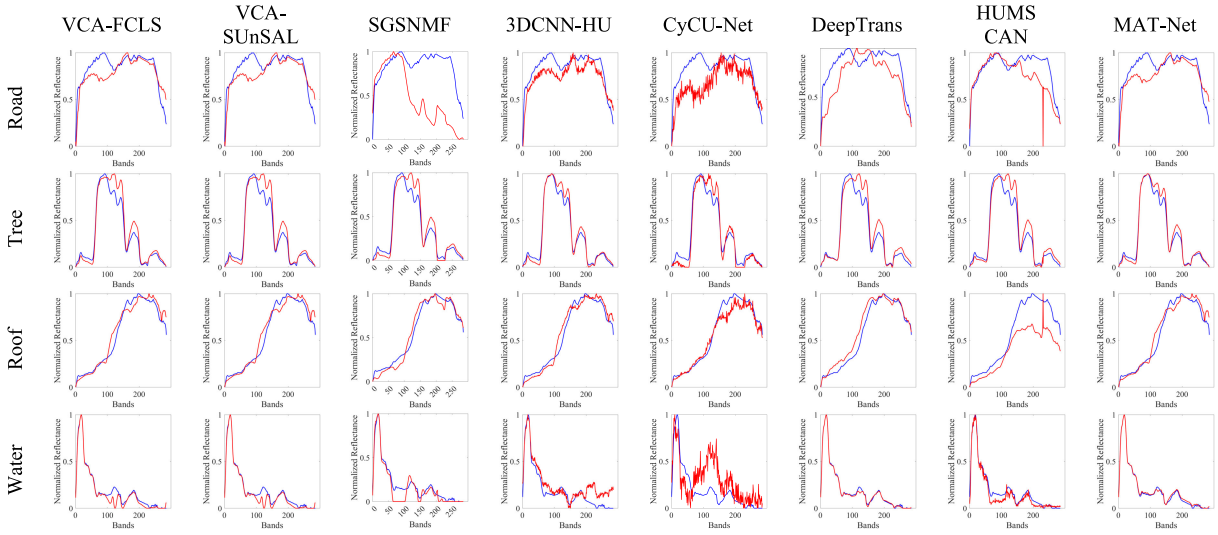



Fig. 10. Visual comparison of the endmembers obtained by different methods on the Apex dataset. Red line: estimated endmembers and blue line: reference endmembers.

water abundance and do not provide accurate road abundance estimations.

Similarly, SUnSAL, CyCU-Net, and HUMSCAN exhibit overestimations of water abundance across numerous tree and road pixels.

Fig. 10 shows the estimated endmembers. While 3DCNN-HU achieves the best results for the roof endmember category, the proposed method demonstrates superior overall performance in endmember estimation. Additionally, the endmember spectra estimated by CyCU-Net lack smoothness, and the

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TABLE VI

QUANTITATIVE COMPARISONS ON THE URBAN DATASET. THE BEST PERFORMANCES ARE SHOWN IN BOLD, AND THE SECOND BEST PERFORMANCES ARE SHOWN WITH AN UNDERLINE

| Algorithms | | VCA-FCLS | VCA-SUnSAL | SGSNMF | 3DCNN-HU | CyCU-Net | DeepTrans | HUMSCAN | MAT-Net |
|---|---|---|---|---|---|---|---|---|---|
| SAD | Rood | 0.1480 | 0.1480 | 0.1712 | 0.2736 | 0.4527 | 0.1323 | 0.0895 | **0.0644** |
| | Grass | 0.3191 | 0.2549 | 0.0817 | 0.3227 | 0.3801 | 0.1510 | 0.2077 | **0.0655** |
| | Tree | 0.1259 | 0.1227 | 0.1560 | 0.1601 | 0.1354 | 0.2109 | 0.0615 | **0.0527** |
| | Roof | 0.1437 | 0.1437 | 0.2742 | 0.0751 | 0.0628 | 0.2468 | 0.0900 | **0.0626** |
| | Dirt | 0.5432 | 0.5432 | 0.9607 | 0.1538 | 0.0781 | 0.1303 | **0.0440** | 0.0560 |
| | Mean | 0.2560 | 0.2425 | 0.3288 | 0.1970 | 0.2218 | 0.1742 | 0.0985 | **0.0602** |
| RMSE | Rood | 0.3542 | 0.1563 | 0.2983 | 0.2960 | 0.3250 | 0.1968 | 0.1662 | **0.1413** |
| | Grass | 0.5894 | 0.2132 | 0.1286 | 0.3411 | 0.4020 | 0.1734 | 0.1939 | **0.1026** |
| | Tree | 0.3029 | 0.1208 | 0.2469 | 0.2593 | 0.3473 | 0.1318 | 0.1703 | **0.0881** |
| | Roof | 0.2117 | 0.1691 | 0.1900 | 0.1830 | 0.1813 | 0.1338 | 0.1066 | **0.0600** |
| | Dirt | 0.2668 | 0.2877 | 0.4574 | 0.1563 | 0.2624 | 0.1380 | 0.1726 | **0.1256** |
| | Mean | 0.3450 | 0.1894 | 0.2642 | 0.2471 | 0.3036 | 0.1548 | 0.1619 | **0.1035** |

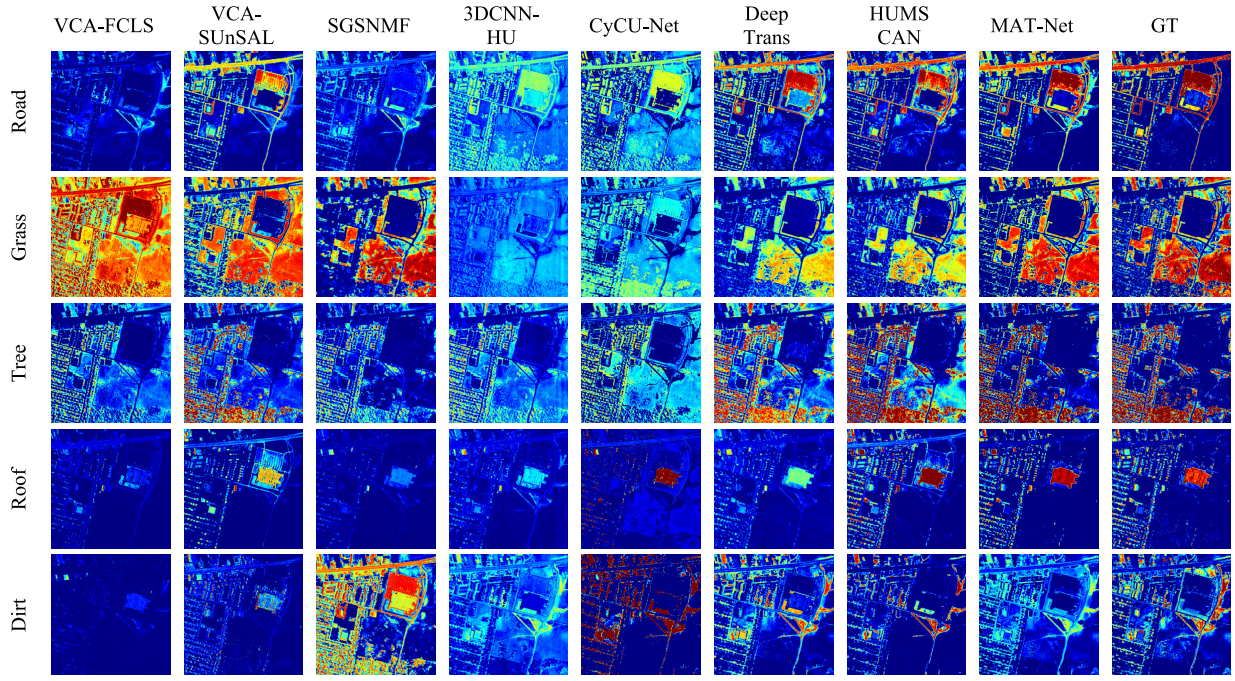



Fig. 11. Visual comparison of the abundances obtained by different methods on the Urban dataset.

road and roof endmember spectra estimated by HUMSCAN exhibit notable outliers at certain bands where reflectance experiences significant fluctuations. Notably, SGSNMF fails to correctly extract the road endmember, as shown in Fig. 10. While 3DCNN-HU accurately estimates the road endmember, it significantly misestimates the water endmember. This misestimation results in a significant overestimation of water abundance within the road pixels, as illustrated in Fig. 9.

*5) Urban Dataset:* Table VI quantitatively presents the performance evaluation results of all methods on the Urban dataset. MAT-Net demonstrates superior performance, achieving the lowest mean values for both SAD and RMSE. DeepTrans and HUMSCAN follow MAT-Net in performance. Traditional methods struggle with accurately extracting the endmember spectra for dirt.

Fig. 11 presents the visualization of abundance estimation results for the Urban dataset. Compared to other methods, MAT-Net achieves the best visual effect in abundance maps, though it slightly underestimates the abundance of dirt. The abundance maps produced by FCLS, SGSNMF, 3DCNN-HU, and CyCU-Net exhibit significant deviations from the reference abundance maps. Influenced by incorrectly extracted dirt endmembers, SGSNMF fails to accurately capture the abundance of roads. DeepTrans falls short in estimating the abundance of grass and roof. While it accurately depicts the main parts of the dirt, it overestimates the abundance of dirt in some road-dominated pixels. HUMSCAN underrepresents the abundance of grass, and its dirt abundance map lacks detailed texture.

Fig. 12 presents the estimated endmembers. The endmember spectra extracted by MAT-Net, HUMSCAN, and DeepTrans are the closest to the reference endmember spectra, successfully capturing detailed spectral features of the road, roof, and dirt. In contrast, the road and roof endmembers extracted by 3DCNN-HU and CyCU-Net exhibit abnormal fluctuations.

## IV. DISCUSSION

A comprehensive analysis of the quantitative evaluation results indicates that MAT-Net's unmixing performance is

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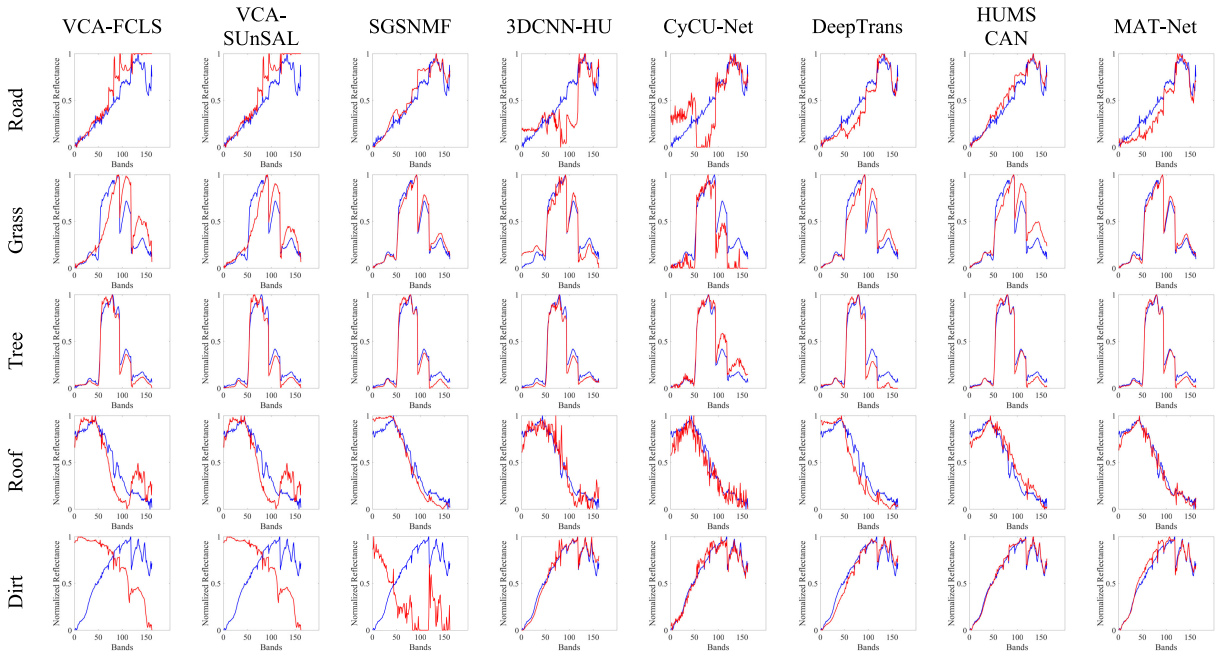



Fig. 12. Visual comparison of the endmembers obtained by different methods on the Urban dataset. Red line: estimated endmembers and blue line: reference endmembers.

significantly superior to that of other comparative algorithms. While SGSNMF demonstrates the highest unmixing precision on the simulated dataset, its performance on real HSIs is poor. SGSNMF performs well on simulated data due to the simpler spatial distribution and Gaussian noise, allowing effective use of the average spectrum of superpixels for generating nearly noise-free initial endmembers. However, in real datasets, complex spatial heterogeneity and more intricate noise types hinder SGSNMF's performance. In visual comparisons across four datasets, MAT-Net consistently achieves the best visualization results among all evaluated methods.

The superior performance of MAT-Net can be attributed to its effective extraction and integration of multiscale information, a capability particularly demonstrated in the unmixing results for soil regions. Accurately estimating the endmember and abundance of soil is challenging, attributed to the intricate spatial distribution characterized by varying plot sizes and uneven dispersion across the image. For the Samson dataset, MAT-Net is the only DL-based method capable of accurately estimating the abundance of small-area soil regions outside the main areas. Similarly, for the Jasper Ridge dataset, MAT-Net produces soil abundance map that most closely align with the reference, underscoring its effectiveness in harnessing multiscale information.

For the Apex dataset, which lacks reference endmembers and abundances, the abundance maps obtained by unmixing with SUnSAL, based on manually extracted endmembers, are prone to errors. However, the water abundance map obtained by MAT-Net does not contain the obvious erroneous redundant textures present in the land areas. This accuracy improvement is due to the transformer encoder's ability to finely adjust the weights of small-scale spatial features corresponding to redundant textures and large-scale spatial features corresponding to water bodies during feature fusion.

To further evaluate the effectiveness of the modules incorporated in MAT-Net, ablation experiments and comparisons of processing times were performed on various datasets, including a simulated dataset with 20-dB noise, as well as the Samson, Jasper Ridge, Apex, and Urban datasets.

### A. Ablation Experiment

Our ablation study focuses on the modules and strategies used to enhance the existing dual-stream unmixing network frameworks. These enhancements include a block-by-block branching strategy, a multibranch architecture, and a transformer encoder. The settings for the ablation experiments are detailed below: when the block-by-block branching strategy is omitted, the spatial stream of the CNN encoder transitions to a configuration with three independent branches. In the absence of the multibranch architecture, the spatial stream of the CNN encoder simplifies to a single branch with convolutional kernels of sizes $5 \times 5$, $3 \times 3$, and $1 \times 1$. If the transformer encoder is excluded, the multiscale features are fused using a $1 \times 1$ convolutional layer before being introduced to the decoder, thereby altering the feature fusion process.

Table VII presents the results of the ablation experiments. As shown in Table VII, MAT-Net performs best when it includes all modules and strategies. The exclusion of the block-by-block branching strategy has a relatively minor impact on network performance. Integrating only the multibranch architecture bolsters MAT-Net's ability to perceive spatial multiscale features. Nonetheless, the absence of effective feature fusion methods curtails its potential to enhance the network's unmixing performance. Likewise, the addition of a transformer encoder alone to fuse spectral features and single-scale spatial features yields only a slight improvement in unmixing accuracy. This underscores that the multibranch architecture, and the transformer encoder are synergistic in

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TABLE VII
QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS

| Module | | | Simulated dataset (20dB) | | Samson | | Jasper Ridge | | Apex | | Urban | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Block-by block branching strategy | Multi-branch architecture | Transformer encoder | SAD | RMSE | SAD | RMSE | SAD | RMSE | SAD | RMSE | SAD | RMSE |
| √ | √ | √ | **0.0474** | **0.0256** | **0.0276** | **0.0546** | **0.0568** | **0.0827** | **0.0787** | **0.1101** | **0.0602** | **0.1035** |
| √ | √ | × | 0.0478 | 0.0584 | 0.0359 | 0.0855 | 0.1292 | 0.1149 | 0.0791 | 0.1749 | 0.0766 | 0.1259 |
| × | √ | √ | 0.0527 | 0.0419 | 0.0308 | 0.0699 | 0.1186 | 0.1061 | 0.0794 | 0.1381 | 0.0635 | 0.1230 |
| × | × | √ | 0.0523 | 0.0520 | 0.0361 | 0.1080 | 0.0956 | 0.1428 | 0.0788 | 0.1663 | 0.0635 | 0.1248 |
| × | × | × | 0.0535 | 0.0661 | 0.0693 | 0.1171 | 0.1889 | 0.1202 | 0.0820 | 0.1771 | 0.0745 | 0.1393 |

TABLE VIII
COMPUTATIONAL TIMES (IN SECONDS) FOR DIFFERENT DATASETS

| Method | Simulated | Samson | Jasper | Apex | Urban |
|---|---|---|---|---|---|
| FCLS | 0.40 | 0.41 | 0.71 | 0.74 | 8.77 |
| SUnSAL | 0.24 | 0.24 | 0.39 | 0.49 | 12.47 |
| SGSNMF | 3.26 | 7.35 | 10.04 | 12.07 | 103.79 |
| 3DCNN-HU | 42.96 | 172.69 | 248.04 | 210.48 | 428.53 |
| CyCU-Net | 9.13 | 15.94 | 6.32 | 23.05 | 129.13 |
| DeepTrans | 9.21 | 13.21 | 14.88 | 18.99 | 120.96 |
| HUMSCAN | 94.17 | 46.45 | 67.21 | 157.31 | 8365.82 |
| MAT-Net | 14.37 | 15.00 | 15.27 | 20.82 | 189.91 |
| MAT-Net (w/o $B^4$) | 18.78 | 18.59 | 20.58 | 25.43 | 237.28 |

their design contribution. Furthermore, network performance is notably compromised when both the multibranch architecture, and the transformer encoder are excluded.

### B. Processing Time

Table VIII shows the running time of all the compared methods. All tests are conducted on a computer with Intel Core i7-12700H CPU (2.30 GHZ), 16 GB RAM, and GeForce RTX 3060 GPU. Most methods experience longer computational times when applied to more complex datasets. Traditional methods exhibit high-computational efficiency, maintaining lower processing times. The two joint spectral-spatial networks compared in the study, 3DCNN-HU and HUMSCAN, exhibit the longest computational times due to their intricate network architectures and the extensive number of parameters they entail. Among the DL-based approaches, the proposed method demonstrates competitive computational efficiency.

To verify the effectiveness of the proposed block-by-block branching strategy in reducing the computational cost, we removed it from MAT-Net (excluding $B^4$) as in the ablation experiments and recorded the processing time in the last row of Table VIII. From Table VIII, it can be observed that the block-by-block branching strategy has reduced the processing time by approximately one-fifth.

## V. CONCLUSION

This article introduces MAT-Net, a MAT-Net designed for HU, which utilizes a novel combination of CNNs and transformers. MAT-Net features a dual-stream, multibranch CNN encoder for extracting multiscale features and incorporates a transformer encoder to enable effective feature fusion. By adeptly merging spectral and multiscale spatial information, MAT-Net significantly enhances HU performance. Comparative analyses across various datasets demonstrate that MAT-Net surpasses competing methods. Considering the high-dimensional spectral information in HSIs and the complex shapes of ground objects, $a$-direction for future research involves integrating spectral attention mechanisms and deformable convolutions within the CNN encoder to further refine the extraction of spectral-spatial features.

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